

Lattice Monte Carlo Foundations

The Two-Dimensional Ising Model and Pure-Gauge Calculations

QCD Prediction Project

Abstract

This document explains the mathematical operations in two related lattice Monte Carlo tracks. The two-dimensional Ising model supplies a small, exactly enumerable test of local Metropolis updates, thermalization, and correlated measurements. The pure-gauge track then applies the same numerical ideas to compact U(1), pure SU(2), and pure SU(3) models with periodic boundaries and the Wilson gauge action. Each expression is connected to the data it consumes, the quantity it produces, and its algorithmic role. Dynamical fermions are outside this calculation.

Contents

1	Calculation at a glance	2
2	Notation	3
3	Step 1: construct the lattice	3
3.1	Allowed sites and volume	3
3.2	Flattened storage	3
3.3	Periodic neighbors	4
4	Toy-model track: the two-dimensional Ising model	4
4.1	Hamiltonian and bond convention	4
4.2	Local spin-flip energy	5
4.3	Metropolis update and sweep definition	5
4.4	Measurements	6
4.5	Exact 2×2 enumeration	6
4.6	Ising validation gates	7
4.7	Relationship to the gauge track	7
5	Step 2: initialize the gauge field	7
6	Step 3: form plaquettes and the Wilson action	8
6.1	Oriented plaquette	8
6.2	Wilson gauge action	8
7	Step 4: update one gauge link	8
7.1	Proposal	8
7.2	Local action difference	9
7.3	Metropolis decision	9

8 Step 5: thermalize and measure	9
8.1 Sampling schedule	9
8.2 Mean plaquette	9
8.3 Wilson loops	10
9 Step 6: extract the static potential	10
10 Step 7: account for correlated samples	10
11 Step 8: validation gates	11
12 Reference algorithm	12
13 How to interpret the final gauge result	12

1 Calculation at a glance

Both tracks begin with the same validated lattice geometry, deterministic random streams, sampling schedule, and statistical analysis. The Ising track stores spins on sites and is the first end-to-end validation problem. The gauge track stores group elements on links and has the following dependency chain:

1. Convert lattice extents into site indices and periodic neighbors.
2. Allocate and initialize one gauge link per site and direction.
3. Form plaquettes and evaluate the Wilson gauge action.
4. Update links with a Markov-chain Monte Carlo algorithm.
5. Discard thermalization, then measure plaquettes and Wilson loops.
6. Estimate autocorrelation-aware statistical uncertainties.
7. Convert Wilson-loop ratios into an effective static potential.
8. Validate structural identities and compare independent chains.

The output of one stage is the input of the next:

Stage	Consumes	Produces
Geometry	Extents and boundaries	Indices, strides, neighbors
Gauge field	Geometry and initial-state policy	Links $U_\mu(x)$
Action	Links and coupling β	Plaquettes and $S_W[U]$
Update	Current field and random numbers	Next Markov-chain state
Measurement	Thermalized field samples	$\bar{P}$ and $W(r, t)$
Statistics	Correlated measurement history	Means, covariance, errors
Potential	Correlated Wilson loops	$aV_{\text{eff}}(r, t)$, then $V(r)$

2 Notation

Symbol	Meaning
D	Number of Euclidean lattice dimensions.
$x = (x_0, \dots, x_{D-1})$	Integer coordinate of one lattice site.
L_μ	Number of sites in direction μ .
$\mathcal{V}$	Total number of lattice sites. A calligraphic symbol distinguishes volume from the potential $V(r)$.
μ, ν	Direction indices in $\{0, \dots, D-1\}$.
$\hat{\mu}$	Unit displacement in direction μ .
$U_\mu(x)$	Gauge link from x to $x + \hat{\mu}$.
G	Gauge group: initially $U(1)$, then $SU(2)$ or $SU(3)$.
N	Matrix dimension for $SU(N)$; QCD uses $N = 3$.
s_x	Ising spin at site x , with $s_x \in \{-1, +1\}$.
J	Ising nearest-neighbor interaction strength.
h	Ising external magnetic field; the first implementation uses $h = 0$.
β_{th}	Thermodynamic inverse temperature $1/(k_B T)$.
β	Bare gauge coupling parameter appearing inside the Wilson action.
a	Lattice spacing; aV is a dimensionless potential.
r, t	Spatial and Euclidean-time extents in lattice units.
N_{meas}	Number of saved measurements in a chain.

3 Step 1: construct the lattice

3.1 Allowed sites and volume

For periodic extents $L_0, \dots, L_{D-1}$, the set of sites is

$$\Lambda = \{x = (x_0, \dots, x_{D-1}) \mid 0 \leq x_\mu < L_\mu\}. \quad (1)$$

Interpretation. Each component x_μ is an integer. Equation (1) defines which coordinates are legal before any fields are allocated.

The total number of sites is

$$\mathcal{V} = |\Lambda| = \prod_{\mu=0}^{D-1} L_\mu. \quad (2)$$

Interpretation. The program must check that every $L_\mu > 0$ and that the product in Equation (2) does not overflow the index type. $\mathcal{V}$ determines the site allocation size.

3.2 Flattened storage

For row-major storage, define strides and the flattened site index by

$$s_0 = 1, \quad s_\mu = \prod_{\nu=0}^{\mu-1} L_\nu, \quad i(x) = \sum_{\mu=0}^{D-1} x_\mu s_\mu. \quad (3)$$

Interpretation. s_μ is the number of contiguous entries skipped when x_μ increases by one. The result $i(x)$ lies in $0 \leq i(x) < \mathcal{V}$ and is used to address site-owned data.

Because every site owns one outgoing link in each direction, a convenient link index is

$$\ell(x, \mu) = D i(x) + \mu, \quad 0 \leq \ell < D\mathcal{V}. \quad (4)$$

Interpretation. Equation (4) maps a site and direction to one flat gauge-field allocation containing $D\mathcal{V}$ group elements.

3.3 Periodic neighbors

Forward and backward neighbors change only coordinate μ :

$$\begin{aligned} (x + \hat{\mu})_\nu &= \begin{cases} (x_\mu + 1) \bmod L_\mu, & \nu = \mu, \\ x_\nu, & \nu \neq \mu, \end{cases} \\ (x - \hat{\mu})_\nu &= \begin{cases} (x_\mu + L_\mu - 1) \bmod L_\mu, & \nu = \mu, \\ x_\nu, & \nu \neq \mu. \end{cases} \end{aligned} \quad (5)$$

Interpretation. The added L_μ in the backward expression keeps the dividend nonnegative before the modulo operation. These neighbors are reused in plaquettes, staples, Wilson paths, and boundary tests.

Step output. A validated lattice shape, a coordinate/index round trip, and forward/backward neighbor operations.

4 Toy-model track: the two-dimensional Ising model

The Ising model is not a gauge theory and is not itself QCD. It is a compact test problem for the same Markov-chain, random-number, measurement, and error analysis machinery needed later. The first implementation uses a periodic $L_x \times L_y$ square lattice and one spin

$$s_x \in \{-1, +1\} \quad (6)$$

at every site.

4.1 Hamiltonian and bond convention

Count each positive-axis bond once and define

$$E[s] = -J \sum_{x \in \Lambda} [s_x s_{x+\hat{0}} + s_x s_{x+\hat{1}}] - h \sum_{x \in \Lambda} s_x. \quad (7)$$

For positive J , aligned neighboring spins lower the energy. Equation (7) contains $2\mathcal{V}$ bond terms, and periodic neighbors are evaluated with Equation (5).

On a 2×2 periodic lattice, the forward and backward neighbor in one axis have the same coordinate. They must nevertheless appear twice in the local-neighbor multiset because they represent two periodic bond occurrences. Removing the duplicate changes the model and invalidates the exact reference values below.

The equilibrium probability of a configuration is

$$\pi[s] = \frac{1}{Z} \exp[-\beta_{\text{th}} E[s]], \quad Z = \sum_{\{s\}} \exp[-\beta_{\text{th}} E[s]]. \quad (8)$$

It is often convenient to use the dimensionless coupling $K = \beta_{\text{th}} J$. Here $\beta_{\text{th}} = 1/(k_{\text{B}} T)$ is an inverse temperature. It is conceptually different from the bare gauge parameter β in the Wilson action.

4.2 Local spin-flip energy

Propose changing only the spin at site x :

$$s'_x = -s_x. \quad (9)$$

Every term not containing s_x cancels in $\Delta E = E[s'] - E[s]$. If $\mathcal{N}(x)$ is the four-entry multiset containing the forward and backward neighbor in both axes, the affected energy before the flip is

$$E_{\text{old},x} = -Js_x \sum_{y \in \mathcal{N}(x)} s_y - hs_x, \quad (10)$$

and after the flip it is

$$E_{\text{new},x} = +Js_x \sum_{y \in \mathcal{N}(x)} s_y + hs_x. \quad (11)$$

Subtracting gives the local formula

$$\boxed{\Delta E = 2s_x \left(J \sum_{y \in \mathcal{N}(x)} s_y + h \right)}. \quad (12)$$

This formula should be tested against a deliberately slow calculation of the full energy before and after every possible spin flip on small lattices.

4.3 Metropolis update and sweep definition

For a symmetric flip proposal, accept with probability

$$P_{\text{accept}} = \min(1, \exp[-\beta_{\text{th}} \Delta E]). \quad (13)$$

For $u \sim \text{Uniform}[0, 1)$, the numerically stable decision is

$$\log u < -\beta_{\text{th}} \Delta E. \quad (14)$$

When $\Delta E \leq 0$, accept without evaluating the logarithm. One sweep is exactly $\mathcal{V}$ attempted spin flips. The program must record whether those attempts use a fixed site order, a new random permutation each sweep, or random sites sampled with replacement; these policies are not identical finite-chain procedures even though each can target the same equilibrium distribution.

A minimal run is

```
initialize all spins and a deterministic random stream
```

```
repeat N_warmup times:
```

```
    attempt one sweep of local spin flips
```

```
repeat N_measurements times:
```

```
    repeat N_spacing times:
```

```
        attempt one sweep of local spin flips
```

```
        record energy, magnetization, acceptance count, and sweep number
```

Warm-up length is not proven adequate merely because it was configured. Compare ordered and random initial states, inspect histories, and repeat with a longer warm-up and measurement spacing.

4.4 Measurements

For $\mathcal{V} = L_x L_y$, define total and per-site energy by

$$E = E[s], \quad e = \frac{E}{\mathcal{V}}, \quad (15)$$

and total and per-site magnetization by

$$M = \sum_{x \in \Lambda} s_x, \quad m = \frac{M}{\mathcal{V}}. \quad (16)$$

At zero field and finite volume, spin-flip symmetry gives $\langle m \rangle = 0$ when both magnetized sectors are sampled. Therefore record m , $|m|$, and m^2 , rather than interpreting a near-zero mean magnetization as an absence of ordering. Also record the acceptance fraction

$$f_{\text{accept}} = \frac{N_{\text{accepted}}}{N_{\text{attempted}}}. \quad (17)$$

Measurements from consecutive sweeps are correlated and must use the autocorrelation or blocking treatment in Step 7.

4.5 Exact 2×2 enumeration

For $h = 0$, enumerate all $2^{\mathcal{V}} = 16$ configurations of a 2×2 lattice. With the bond convention in Equation (7), the energy levels and degeneracies are

Energy	Degeneracy	Configuration class
$-8J$	2	all spins aligned
0	12	one-versus-three or adjacent two-versus-two
$+8J$	2	checkerboard

Consequently,

$$Z_{2 \times 2} = 2e^{8K} + 12 + 2e^{-8K} = 4[\cosh(8K) + 3], \quad (18)$$

and the exact per-site energy is

$$\langle e \rangle_{2 \times 2} = -2J \frac{\sinh(8K)}{\cosh(8K) + 3}. \quad (19)$$

The exact absolute magnetization per site is

$$\langle |m| \rangle_{2 \times 2} = \frac{e^{8K} + 2}{2[\cosh(8K) + 3]}, \quad \langle m \rangle_{2 \times 2} = 0. \quad (20)$$

At $K = 0$, these expressions give $\langle e \rangle = 0$ and $\langle |m| \rangle = 3/8$. As $K \rightarrow \infty$, they approach $\langle e \rangle = -2J$ and $\langle |m| \rangle = 1$.

The enumerator should construct configurations independently of the Monte Carlo code, evaluate Equation (7) directly, and accumulate weighted observables using Equation (8). It is a test oracle, not a production sampling algorithm.

4.6 Ising validation gates

Before using the toy simulation as evidence that the sampling infrastructure works, require all of the following:

1. every local ΔE equals a full-energy recomputation;
2. the enumerator reproduces the degeneracies and exact formulas above;
3. the same seed and update policy reproduce the same CPU history;
4. ordered and random starts agree after adequate warm-up;
5. Monte Carlo estimates agree with exact 2×2 values within autocorrelation-aware uncertainty;
6. uncertainty grows when fewer effectively independent samples are used; and
7. measurements retain chain identity and sweep number.

4.7 Relationship to the gauge track

Concept	Ising track	Pure-gauge track
Degrees of freedom	Spins s_x on sites	Group elements $U_\mu(x)$ on links
Statistical weight	$\exp[-\beta_{\text{th}} E]$	$\exp[-S_W]$, with β inside S_W
Local change	ΔE from neighboring spins	ΔS from neighboring plaquettes
First observables	$e, m, m $	Plaquette and Wilson loops
Shared machinery	Geometry, RNG, sweeps, histories, autocorrelation, uncertainty	Same

5 Step 2: initialize the gauge field

A directed link stores an element of the chosen gauge group:

$$U_\mu(x) \in G, \quad G \in \{\text{U}(1), \text{SU}(2), \text{SU}(3)\}. \quad (21)$$

For an $\text{SU}(N)$ link, group membership requires

$$U_\mu^\dagger(x) U_\mu(x) = I, \quad \det U_\mu(x) = 1. \quad (22)$$

Interpretation. $U_\mu(x)$ transports a color vector from x to $x + \hat{\mu}$. Traversing the link backward uses $U_\mu^\dagger(x)$. The identities in Equation (22) are also inexpensive numerical diagnostics.

A cold start uses $U_\mu(x) = I$. A hot start draws a random valid group element for each link. Both starts should approach the same equilibrium distribution after sufficient thermalization.

Step output. A gauge field containing DV valid link values and reproducibly seeded random-number streams.

6 Step 3: form plaquettes and the Wilson action

6.1 Oriented plaquette

The elementary loop in the positive μ - ν orientation is

$$U_{\mu\nu}(x) = U_\mu(x) U_\nu(x + \hat{\mu}) U_\mu^\dagger(x + \hat{\nu}) U_\nu^\dagger(x). \quad (23)$$

Interpretation. Read Equation (23) from left to right as four path segments: forward in μ , forward in ν , backward in μ , and backward in ν . The path returns to x . Its traced value is gauge invariant even though an individual link is not.

6.2 Wilson gauge action

For pure $SU(N)$, use

$$S_W[U] = \beta \sum_x \sum_{\mu < \nu} \left(1 - \frac{1}{N} \text{Re Tr } U_{\mu\nu}(x) \right). \quad (24)$$

Interpretation. The inner sum counts every unoriented plaquette once. A plaquette close to the identity has a small contribution; rough field configurations have a larger action. The convention in Equation (24) must be used consistently by the global action, local update, and observables.

The corresponding ensemble expectation of an observable $\mathcal{O}$ is

$$\langle \mathcal{O} \rangle = \frac{1}{Z} \int \mathcal{D}U \mathcal{O}[U] e^{-S_W[U]}, \quad Z = \int \mathcal{D}U e^{-S_W[U]}. \quad (25)$$

Interpretation. $\mathcal{D}U$ means integration over every gauge link using the invariant group measure. Z normalizes the probability distribution. The program does not enumerate this integral; it estimates it with correlated samples from a Markov chain whose target weight is proportional to $e^{-S_W[U]}$.

Step output. A global action definition, a local plaquette primitive, and a target probability distribution for the sampler.

7 Step 4: update one gauge link

7.1 Proposal

Draw a small group-valued perturbation $R \in G$ and propose

$$U'_\mu(x) = R U_\mu(x). \quad (26)$$

Interpretation. Because both factors belong to G , the proposal remains in G . The proposal distribution should be symmetric, or its forward/reverse probability ratio must be included in a more general Metropolis–Hastings test.

7.2 Local action difference

Only the $2(D-1)$ plaquettes containing $U_\mu(x)$ can change. If $\mathcal{P}(x, \mu)$ denotes that set, compute

$$\Delta S = S_{\text{local}}[U'] - S_{\text{local}}[U], \quad (27)$$

$$S_{\text{local}}[U] = \beta \sum_{p \in \mathcal{P}(x, \mu)} \left(1 - \frac{1}{N} \text{Re Tr } U_p \right). \quad (28)$$

Interpretation. Equations (27)–(28) replace an $O(D^2\mathcal{V})$ full-action recomputation with a fixed-size local calculation. As a correctness test, this ΔS must agree with the difference of two slow global-action evaluations.

7.3 Metropolis decision

The acceptance probability is

$$P_{\text{accept}} = \min(1, e^{-\Delta S}). \quad (29)$$

For $u \sim \text{Uniform}[0, 1)$, accept if

$$u < P_{\text{accept}}, \quad \text{equivalently} \quad \log u < -\Delta S. \quad (30)$$

Interpretation. If $\Delta S \leq 0$, the proposal is always accepted. If $\Delta S > 0$, it is accepted with probability $e^{-\Delta S}$. The logarithmic form in Equation (30) avoids numerical underflow. Visiting every site x and direction μ once constitutes one sweep.

Step output. One new Markov-chain state plus acceptance counters used to diagnose proposal size and sampling behavior.

8 Step 5: thermalize and measure

8.1 Sampling schedule

First discard N_{warmup} sweeps. Then save one measurement after every N_{spacing} sweeps until N_{meas} measurements have been recorded. Saving less frequently can reduce storage and correlation, but it does not substitute for estimating autocorrelation.

8.2 Mean plaquette

There are

$$N_{\text{plaq}} = \mathcal{V} \frac{D(D-1)}{2} \quad (31)$$

unoriented plaquettes on a periodic isotropic lattice. Their normalized mean is

$$\bar{P} = \frac{1}{N_{\text{plaq}}} \sum_x \sum_{\mu < \nu} \frac{1}{N} \text{Re Tr } U_{\mu\nu}(x). \quad (32)$$

Interpretation. $\bar{P}$ is inexpensive and is the first physical sanity check. Its history helps reveal thermalization and frozen or unstable updates, but it does not by itself determine the static potential.

8.3 Wilson loops

For a rectangular closed path $C_{r,t}$, define

$$W(r, t) = \frac{1}{N} \left\langle \text{Re Tr } \mathcal{P} \prod_{(x, \mu) \in C_{r,t}} U_\mu(x) \right\rangle. \quad (33)$$

Interpretation. $\mathcal{P}$ means path ordering. Each forward segment contributes a link $U_\mu(x)$; each backward segment contributes its dagger. The trace closes the color indices and makes the result gauge invariant. Measure several spatial widths r and time extents t on every saved configuration.

Step output. Correlated histories $\bar{P}_k$ and $W_k(r, t)$, each stored with chain ID and sweep number.

9 Step 6: extract the static potential

A spectral decomposition has the schematic form

$$W(r, t) = \sum_n |c_n(r)|^2 e^{-aE_n(r)t}. \quad (34)$$

Interpretation. The chosen loop operator overlaps with several energy eigenstates. Excited states decay faster with t , but the statistical signal also becomes noisier. This creates the practical need for a fit window.

At sufficiently large t , the lowest state dominates:

$$W(r, t) \simeq A(r) e^{-aV(r)t}. \quad (35)$$

Interpretation. $V(r)$ is the static-source potential and $A(r)$ is an unknown overlap factor. The product $aV(r)$ is dimensionless when t is counted in lattice time steps.

The neighboring-time ratio removes $A(r)$:

$$aV_{\text{eff}}(r, t) = \log \left(\frac{W(r, t)}{W(r, t+1)} \right). \quad (36)$$

Interpretation. Plot Equation (36) against t at fixed r . A region consistent with a constant is a candidate ground-state plateau. Fit that region to obtain $aV(r)$, and vary the fit window to estimate a systematic uncertainty.

Step output. An effective-potential curve for each r , a documented plateau window, and a fitted $aV(r)$ with uncertainty.

10 Step 7: account for correlated samples

For a measurement history $O_1, \dots, O_{N_{\text{meas}}}$, first compute

$$\bar{O} = \frac{1}{N_{\text{meas}}} \sum_{k=1}^{N_{\text{meas}}} O_k. \quad (37)$$

Define the lag- τ autocovariance and normalized autocorrelation by

$$C_O(\tau) = \frac{1}{N_{\text{meas}} - \tau} \sum_{k=1}^{N_{\text{meas}} - \tau} (O_k - \bar{O})(O_{k+\tau} - \bar{O}), \quad (38)$$

$$\rho_O(\tau) = \frac{C_O(\tau)}{C_O(0)}. \quad (39)$$

Interpretation. $\rho_O(\tau)$ measures how strongly samples separated by τ saved measurements remain related. It equals one at zero lag and should decay toward zero. Different observables can have different autocorrelation scales.

With a justified truncation window τ_{max} , estimate

$$\tau_{\text{int}} = \frac{1}{2} + \sum_{\tau=1}^{\tau_{\text{max}}} \rho_O(\tau). \quad (40)$$

Interpretation. The sum cannot be extended indefinitely because large-lag estimates become noise dominated. A windowing rule or a blocking study should justify τ_{max} .

The approximate effective sample count and variance of the mean are

$$N_{\text{eff}} \simeq \frac{N_{\text{meas}}}{2\tau_{\text{int}}}, \quad (41)$$

$$\text{Var}(\bar{O}) \simeq \frac{2\tau_{\text{int}}}{N_{\text{meas}}} \text{Var}(O). \quad (42)$$

Interpretation. If the samples were independent, $\tau_{\text{int}} \approx 1/2$ and $N_{\text{eff}} \approx N_{\text{meas}}$. Positive correlation reduces the effective sample count and enlarges the error bar. Wilson-loop ratios and potential fits must be recomputed inside each block-jackknife or bootstrap resample so their cross-correlations are preserved.

Step output. Autocorrelation estimates, a stable block size, covariance-aware uncertainties, and an effective sample count for each observable.

11 Step 8: validation gates

The following checks should be passed in order:

1. coordinate-to-index round trips and every periodic boundary;
2. unitarity and determinant constraints for generated links;
3. local ΔS versus a recomputed global action difference;
4. invariance of traced closed loops under random gauge transformations;
5. agreement of hot and cold starts after thermalization;
6. agreement of independent chains within correlated uncertainties;
7. a tiny-lattice calculation or published benchmark with matching conventions; and
8. volume and lattice-spacing studies before making physical claims.

A failed gate does not necessarily make the raw run useless for debugging, but it does make the resulting observable scientifically unvalidated.

Step output. A validation record stating which structural, numerical, statistical, and physical checks passed or failed.

12 Reference algorithm

```
validate configuration
construct lattice indices and periodic neighbors
initialize all links and deterministic random streams

repeat N_warmup times:
    sweep over every site x and direction mu

repeat N_measurements times:
    repeat N_spacing times:
        sweep over every site x and direction mu
        measure mean plaquette and Wilson loops W(r,t)

block the correlated measurements
for every jackknife or bootstrap resample:
    recompute W(r,t)
    compute a  $V_{\text{eff}}(r,t) = \log[W(r,t) / W(r,t+1)]$ 
    fit the selected plateau

report means, covariance-aware uncertainty, diagnostics, and provenance
```

13 How to interpret the final gauge result

A reported value of $aV(r)$ is not automatically a continuum QCD prediction. It is a result for the documented action, coupling, volume, boundary conditions, operator, sampling procedure, and fit window. A defensible report therefore includes the resolved configuration, action convention, chain histories, autocorrelation treatment, validation results, and known finite-volume and lattice-spacing limitations.